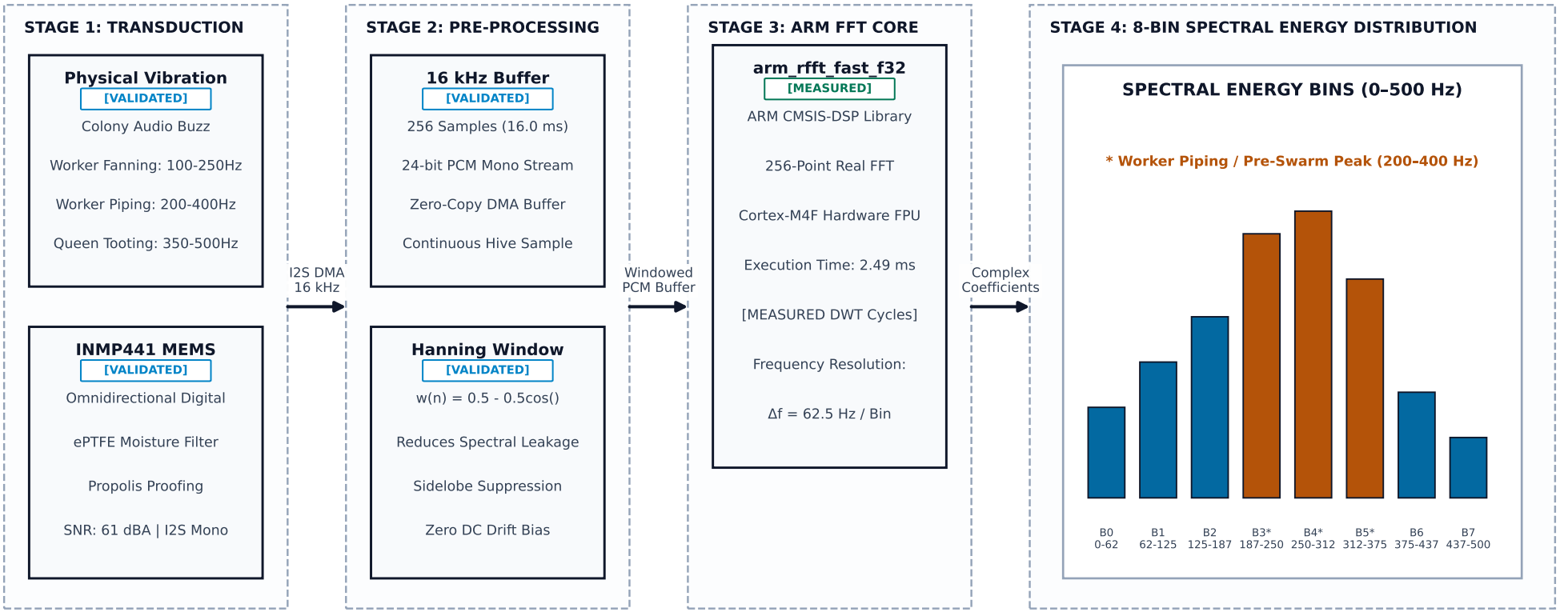
FIGURE 05: BIO-ACOUSTIC DIGITAL SIGNAL PROCESSING (DSP) PIPELINE

From Hive Acoustic Vibrations to 8-Bin Spectral Energy Distribution [CMSIS-DSP arm_rfft_fast_f32, 2.49 ms Latency]



Edge vs Gateway Feature Handoff

[VALIDATED]

On-Node MCU calculates 8 spectral bin energies and packs them into the 33-byte LoRa packet.

Raspberry Pi Gateway Reader feeds these 8 bins + environmental features into Model 2 (Random Forest) for multi-sensor colony classification.

[MEASURED] [VALIDATED] [CALCULATED] [SIMULATED] [DEMONSTRATED]